MAHEDI HASAN

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Career Objectives

My career objective is to establish myself as a distinguished academic and researcher within the next decade by continuously expanding my knowledge with the collaboration of globally recognized researchers and academic professionals. I aim to advance cutting-edge research, produce impactful scholarly contributions and translate them into innovative solutions to technological challenges and foster intellectual growth of future generations through effective teaching and mentorship. I also aspire to apply my expertise to address Bangladesh's societal and technological challenges while contributing to the advancement of higher education, sustainable national development, and global competitiveness.

Education

B.Sc. in Computer Science and Engineering 2018–2023
Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh
CGPA: 3.85/4.00 (Ranked 5th out of 120)
Thesis: Cross-Attention-Based Multimodal Fusion of Visual and Skeletal Features for Robust Human Fall Detection

Research Interests

Computer Vision • Multimodal Learning • Vision Language Model(VLM)• Spatiotemporal Modeling • Video Analytics • Visual Perception • Healthcare AI • Intelligent Transportation Systems

Research Experience

Cross-Attention Visual–Skeletal Feature Fusion for Human Fall Detection | Undergraduate Researcher
Chittagong University of Engineering & Technology | Chittagong, Bangladesh 2023–2024

- Developed a Multimodal Human-Fall Detection framework fusing visual and skeletal features, and a key frames selection algorithm based on body motion to improve the robustness, and efficiency
- Extracted human body keypoints by **Pose Estimation** techniques, followed by modeling and feature extraction using Graph Neural Network(GNN) and visual features are extracted using Vision Transformer(ViT)
- Designed a Cross-attention fusing mechanism to represent and fuse heterogeneous features along with a spatio-temporal model(LSTM) for capturing relationship among frames.
- Evaluated the framework across multiple dataset(UR FALL, Le2i, Multicamera, and datasets) achieving 98%, 95%, and 90% accuracy, with less time and space, presenting research findings through a thesis manuscript, oral defense and a conference presentation, contributing to ongoing research focused on intelligent fall prevention systems

Reproducible Computer Vision Pipeline for Automated Comet Assay Analysis | Supervised Author
International University of Business Agriculture and Technology | Dhaka, Bangladesh 2025–2026

- Developed a Deep Learning framework to automate and analyze DNA-damage quantification from fluorescence microscopy images
- Designed an Attention-guided U-net for DNA head & tail structural segmentation, binary mapping, and morphometric features extraction from microscopic DNA images using Principal Component Analysis (PCA) and connected-component analysis.
- Integrated a Crop-based CNN and Morphometric-only Classifiers for grading and scoring DNA damage as well as validating Validated the extracted measurements against expert-reviewed clinical standards (Bland–Altman Analysis, Pearson Correlation, and Concordance Correlation Coefficient)
- Presented the research findings to the *Egyptian Journal of Medical Human Genetics*, aiming to contribute to early cancer detection, clinical genotoxicology, biomonitoring, and comet assay analysis and applications.

Two-Stage Deep Learning Framework for Video Anomaly Detection | First Author
International University of Business Agriculture and Technology | Dhaka, Bangladesh 2024–2025

- Developed a two-stage single pipeline deep learning framework combining CNN-ViT along with a temporal feature preservation model for for surveillance video anomaly detection and classification
- Designed a hybrid computationally efficient keyframe-extraction algorithm combining frame differencing, HDBSCAN clustering, and brightness-based methods
- Integrated Convolutional Neural Networks to isolate anomalous events and Vision Transformers for multi-class classification of anomaly types (robbery, assault, arson, vandalism) into a unified pipeline
- Evaluated the framework on the UCF-Crime dataset, achieving 98% accuracy for binary anomaly detection and 95% accuracy for multi-class classification;published in (IEEE) and aiming to contribute in intelligent public safety monitoring

Human-in-the-Loop Continual Learning for Adaptive Emotion Recognition | First Author
International University of Business Agriculture and Technology | Dhaka, Bangladesh 2024–2025

- Designed and Experimented a human-in-the-loop (HITL) framework to assess the impact of targeted human intervention on model output on facial emotion recognition.
- Applied a confidence-based intervention strategy to isolate ambiguous predictions (confidence 0.4–0.6) for manual correction to minimize human review to only uncertain cases.
- Integrated a continual (incremental) learning mechanism to update model weights from human feedback without full retraining, enabling adaptive improvement over time.
- Evaluated across four benchmark datasets (FER2013, RAF-DB, AffectNet-7, ExpW), achieving accuracy improvements of 7%, 5%, 10%, and 13% respectively over the baseline Vision Transformer (ViT) model and published in IEEE, contributing to adaptive, and feedback-driven learning systems for deep learning models.

Technical Skills

- **Programming:** Python, C/C++, MATLAB, SQL
- **Frameworks:** PyTorch, TensorFlow/Keras, OpenCV, Scikit-learn, CUDA
- **Computer Vision Competencies:** Detection, Classification, Segmentation, Pose Estimation, surveillance
- **Deep Learning Techniques:** Multimodal Fusion, Attention Mechanisms, Spatiotemporal Modeling, Incremental Learning, Human-in-the-Loop (HiTL)
- **Tools:** Git, LaTeX, Linux, Google Colab, Kaggle, Jupyter

Publications

Journal Articles

[J1] A. H. Emon, N. Jahan, **M. Hasan**, F. A. Jibon, S. Mohammad, and A. H. M. Kamal, "A Reproducible Comet-Assay Pipeline: Segmentation, Morphometrics, and Binary Grading with Auditable Ablations,"(Accepted) *Journal of Digital Health*, manuscript id: DHJ-26-3135, 2026.

Peer-Reviewed Conference Papers

[C1] **M. Hasan**, J. A. Nabin, N. Mia, F. S. Tamim, S. Mohammad, and D. M. Das, "Two-Stage Vision Transformer-Based Framework for Anomaly Detection and Classification in Surveillance Videos," in *Proceedings of the 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, Chittagong, Bangladesh, 2025, pp. 1–6, doi: 10.1109/ECCE64574.2025.11013374

[C2] **M. Hasan**, K. N. Hasan, F. S. Tamim, M. S. Arefin, and A. W. Reza, "Generative Adversarial Networks for Transaction Anomaly Detection: A Synthetic Data Approach," in *Proceedings of the 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, Rangpur, Bangladesh, 2025, pp. 1–6, doi: 10.1109/QPAIN66474.2025.11172003

[C3] **M. Hasan** et al., "Enhancing Face-to-Emotion Recognition with Vision Transformer and Human-in-the-Loop Approach," in *Proceedings of the 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, Rangpur, Bangladesh, 2025, pp. 1–6, doi: 10.1109/QPAIN66474.2025.11171658

[C4] J. Khatun, N. S. Arnika, M. N. Ahmed, F. S. Tamim, **M. Hasan**, and K. Islam, "CLIP–Longformer Fusion for Multimodal Bengali Meme Understanding: Dataset Development and Benchmarking," in *Proceedings of the 28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025, pp. 3675–3680, doi: 10.1109/ICCIT68739.2025.11490332

[C5] N. Mia, J. A. Nabin, S. Mohammad, **M. Hasan**, F. S. Tamim, and D. Mohon Das, "Federated Deep Learning for Cybersecurity and Intrusion Detection in Decentralized Networks," in *Proceedings of the 2025 3rd International Conference on Advancements in Electrical, Electronics, Communication, Computing and Automation (ICAECA)*, Coimbatore, India, 2025, pp. 1–6, doi: 10.1109/ICAECA63854.2025.11012607

[C6] F. Z. Nishat, S. Islam, J. F. Raisa, M. U. Ahmed, and **M. Hasan**, "Comparing Deep Learning Frameworks for Air Quality Index (AQI) Prediction in Bangladesh: Static vs. Year-Wise Forecasting Evaluation," in *Proceedings of the 28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025, pp. 1148–1153, doi: 10.1109/ICCIT68739.2025.11489545

[C7] F. S. Tamim, Z. S. Taheri, **M. Hasan**, M. S. Arefin, and A. W. Reza, "Association Rule Mining for Analyzing User Listening Patterns and Performance," in *Proceedings of the 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, Rangpur, Bangladesh, 2025, pp. 1–6, doi: 10.1109/QPAIN66474.2025.11171825

[C8] N. Tabassum, S. S. Jese, **M. Hasan** and M. Tasnim, "Multi-Architecture Comparative Framework for Bangla Sign Language Classification Using CNN and Transformer Models," *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, pp. 1-6, doi: 10.1109/QPAIN69676.2026.11546199

[C9] P. B. Barua, S. N. Lima and **M. Hasan**, "Enhanced YOLOv8 with Convolutional Block Attention Module (CBAM) for Real-Time Bangladeshi License Plate Detection," *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, pp. 1-6, doi: 10.1109/QPAIN69676.2026.11546280

Conference Chapter

[B1] T. Ahammad, **M. Hasan**, M. Hasan, M. Sabir Hossain, A. Hoque, and M. M. Rashid, "A New Approach to Solve Job Sequencing Problem Using Dynamic Programming with Reduced Time Complexity," in

N. Chaubey, S. Parikh, and K. Amin (Eds.), *Computing Science, Communication and Security* (*COMS2 2020*), Communications in Computer and Information Science, vol. 1235, Springer, Singapore, 2020, doi: 10.1007/978-981-15-6648-6_25

Teaching Experience

Lecturer | Department of Computer Science & Engineering July 2023–Present
International University of Business Agriculture and Technology | Dhaka, Bangladesh

- **Courses Taught:** Data Structures & Algorithms, Data Communications & Computer Networks, Computer Vision & Image Processing, Machine Learning, Data Science, Java Programming, and Computer Architecture.
- **Course Development & Instruction:** Design and deliver lectures, laboratory sessions, assignments, projects, and examinations, integrating theoretical concepts with hands-on programming and problem-solving activities.
- **Student Mentoring:** Supervise undergraduate projects and research activities, provide academic guidance, and support students in developing programming, problem-solving, and technical skills.
- **Assessment & Academic Support:** Evaluate student performance through examinations, laboratory work, assignments, and projects, providing structured feedback to support academic improvement.

AWARDS AND HONORS

- **Miyan Research Institute Publications Award | IUBAT** (Exceptional research productivity) 2026
- **Dean's Merit List Award | CUET** (CGPA 3.85; 5th merit position) 2023
- **Science Olympiad Champion | Inter-College Science Fair, Madaripur** (1st place) 2017
- **Government Merit Scholarship | Education Ministry of Bangladesh** (PSC, JSC & SSC) 2010, 2013, 2016

Service & Leadership

Competitive Programming Coach & Contest Director | IUBAT IT Society July 2023–Present.
International University of Business Agriculture and Technology | Dhaka, Bangladesh

- Mentor undergraduate students in competitive programming, algorithmic problem solving, and computational thinking through structured training sessions and hands-on practice.
- Organize programming workshops and intra-university contests; serve as problem setter and judge, and coordinate student teams for inter-university programming competitions.

Student Advisor | Department of Computer Science & Engineering July 2023–Present.
International University of Business Agriculture and Technology | Dhaka, Bangladesh

- Advise undergraduate students on academic planning, study strategies, course selection, and career development through individual and group advisory sessions.
- Monitor academic progress and provide targeted guidance to students requiring additional academic support.

References

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